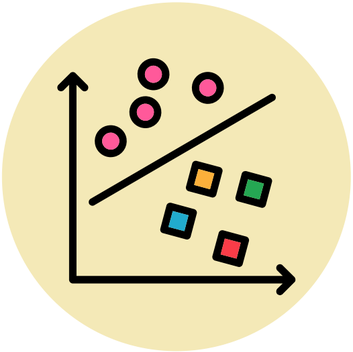

KAZAKH-BRITISH TECHNICAL UNIVERSITY

School of IT and Engineering

# MACHINE LEARNING

## LECTURE #4

Number of credits: 3 (2/0/1)
Course code – CSCI3234
MS Teams team code – **y4ntwlz**

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# MODEL EVALUATION

After building a classification model, the next crucial step is evaluating how well it performs and selecting the best model among alternatives.

Core questions:

- How accurate is my model on unseen data?
- Which metric best represents performance for my problem?
- How do I compare different models reliably?

These questions fall under two topics:

- Model Evaluation – measuring predictive accuracy.
- Model Selection – choosing the best model among candidates.

# MODEL EVALUATION

Model Evaluation is the process of assessing how well a trained machine learning model performs on unseen (test) data.
It measures the model's predictive power, generalization ability, and reliability using quantitative metrics.

In other words, it helps answer the question:
"How accurately and effectively does my model make predictions on new data?"

A model that performs perfectly on training data may fail in real-world scenarios due to overfitting—it memorizes instead of generalizing.
Thus, evaluation must always use unseen test data or validation folds.

Example:
A classifier trained on previous sales predicts whether a new customer will buy a product.
Accuracy on past data ≠ accuracy on future customers.

# MODEL EVALUATION

**Key Points:**

- Goal: To ensure the model performs well not only on training data but also on new, unseen data.
- Data Split: Typically involves separating data into training, validation, and testing sets.
- Metrics Used: Depend on the type of problem:
  - Classification: Accuracy, Precision, Recall, F1-score, ROC-AUC, etc.
  - Regression: Mean Squared Error (MSE), Root Mean Squared Error (RMSE), $R^2$, MAE, etc.

# IMBALANCED DATA CLASSIFICATION

In many real-world tasks, one class occurs much more frequently than the other.

- For example:
- Fraud detection → only 1% of transactions are fraud.
- Disease diagnosis → only few patients have the disease.
- Credit default → most clients pay on time.

In machine learning, **imbalanced data** refers to a situation where the number of observations (samples) in each class is not equal — one class (the majority class) has many more examples than the other (the minority class)

# MODEL EVALUATION

## Confusion Matrix and Terminology

The confusion matrix summarizes predictions versus actual class labels.

| Actual / Predicted | Positive | Negative |
|---|---|---|
| **Positive** (P) | True Positive (TP) | False Negative (FN) |
| **Negative** (N) | False Positive (FP) | True Negative (TN) |

- TP: Correctly predicted positives
- TN: Correctly predicted negatives
- FP: Incorrectly predicted positives
- FN: Missed positives

Total samples = P + N = TP + TN + FP + FN

# MODEL EVALUATION

## Evaluation Metrics for Classifier Performance

*Accuracy and Error Rate*

$$Accuracy = \frac{TP + TN}{P + N}$$

$$Error\ Rate = 1 - Accuracy = \frac{FP + FN}{P + N}$$

Best for: Balanced datasets.
Problem: Misleading in imbalanced data (e.g., detecting rare diseases or fraud).

# MODEL EVALUATION

## Evaluation Metrics for Classifier Performance

*Sensitivity (Recall / True Positive Rate)*

$$Sensitivity = \frac{TP}{TP + FN}$$

**Measures how well the model detects positive cases. Measures completeness—how many actual positives are captured (recall)**

Example (Medical test):
Out of 100 cancer patients, only 30 were correctly identified.

Sensitivity = 30 / 100 = 30%

*Specificity (True Negative Rate)*

$$Specificity = \frac{TN}{TN + FP}$$

**Measures how well the model recognizes negative cases.**

Example:
Out of 9700 non-cancer patients, 9560 were correctly identified.

Specificity = 9560 / 9700 = 98.56%

# MODEL EVALUATION

## Evaluation Metrics for Classifier Performance

*Precision (Positive Predictive Value)*

**Measures exactness—how many predicted positives are actually positive.**

$$Precision = \frac{TP}{TP + FP}$$

Example:
230 predicted positives, only 90 correct → Precision = 90 / 230 = 39.13%

# MODEL EVALUATION

**Evaluation Metrics for Classifier Performance**

*F1 Score (Harmonic Mean)*

**Balances precision and recall.**

$$F1 = 2 \times \frac{Precision \times Recall}{Precision + Recall}$$

*Weighted Fβ Score*

$$F_\beta = (1 + \beta^2) \times \frac{Precision \times Recall}{\beta^2 \times Precision + Recall}$$

β > 1 → More weight on recall
β < 1 → More weight on precision

# MODEL EVALUATION

## Confusion Matrix

| Actual / Predicted | Fraud (1) | Legitimate (0) |
|---|---|---|
| **Fraud (1)** (Positive) | **6 (TP)** | **4 (FN)** |
| **Legitimate (0)** (Negative) | **20 (FP)** | **970 (TN)** |

**TP = 6, FN = 4, FP = 20, TN = 970**

**Accuracy:**

$$\frac{6+970}{1000} = 97.6\%$$

**Precision:**

$$\frac{TP}{TP+FP} = \frac{6}{6+20} = 0.23 = 23\%$$

**Recall:**

$$\frac{TP}{TP+FN} = \frac{6}{6+4} = 0.6 = 60\%$$

**F1-score:**

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} = 2 \times \frac{0.23 \times 0.6}{0.23 + 0.6} = 0.33 = 33\%$$

| Metric | Value | Comment |
|---|---|---|
| Accuracy | 97.60% | High, but misleading |
| Precision | 23% | Many false alarms |
| Recall | 60% | Detects most frauds |
| F1-score | 33% | Balanced indicator |

# MODEL EVALUATION

**Real-world Example**

- Credit Bureau: predicting loan default — default cases are rare but critical.
- Insurance: predicting fraudulent claims — small in number but costly.
- Healthcare: detecting rare diseases — imbalance is natural.

# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

- When evaluating classification models—especially binary classifiers—accuracy alone can be misleading.

- To better understand how well a model separates classes, we use ROC curves and AUC (Area Under Curve).

| Predicted | Actual Positive | Actual Negative |
|---|---|---|
| Positive | True Positive (TP) | False Positive (FP) |
| Negative | False Negative (FN) | True Negative (TN) |

$$\text{TPR (Recall)} = \frac{TP}{TP + FN}$$

$$\text{FPR} = \frac{FP}{FP + TN}$$

# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

- ROC (Receiver Operating Characteristic) curve shows the trade-off between TPR and FPR at various threshold settings.
- The x-axis is the False Positive Rate (FPR).
- The y-axis is the True Positive Rate (TPR).
- Each point on the curve corresponds to a different decision threshold used by the classifie

ROC AUC stands for Receiver Operating Characteristic Area Under the Curve

# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

Suppose a model outputs probabilities for the positive class:

| True Label | Predicted Probability |
|---|---|
| 1 | 0.95 |
| 0 | 0.85 |
| 1 | 0.75 |
| 0 | 0.45 |
| 1 | 0.30 |

1)Sort predictions by their probability scores (descending).
2)For each threshold:
Classify scores ≥ threshold as positive.
Compute TPR and FPR.
3)Plot the points (FPR, TPR) on the graph.
4)Connect them to form the ROC curve.

# MODEL EVALUATION

## ROC Curves and AUC (Area Under the Curve)

- Example Calculation

| Threshold | TP | FP | TN | FN | TPR | FPR |
|---|---|---|---|---|---|---|
| 0.9 | 1 | 0 | 4 | 5 | 0.17 | 0.00 |
| 0.7 | 3 | 1 | 3 | 3 | 0.50 | 0.25 |
| 0.5 | 4 | 2 | 2 | 2 | 0.67 | 0.50 |
| 0.3 | 5 | 3 | 1 | 1 | 0.83 | 0.75 |
| 0.1 | 6 | 4 | 0 | 0 | 1.00 | 1.00 |

# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

AUC (Area Under ROC Curve) represents the probability that a classifier will rank a randomly chosen positive instance higher than a randomly chosen negative instance.

$$AUC = \int_0^1 TPR(FPR)\, d(FPR)$$

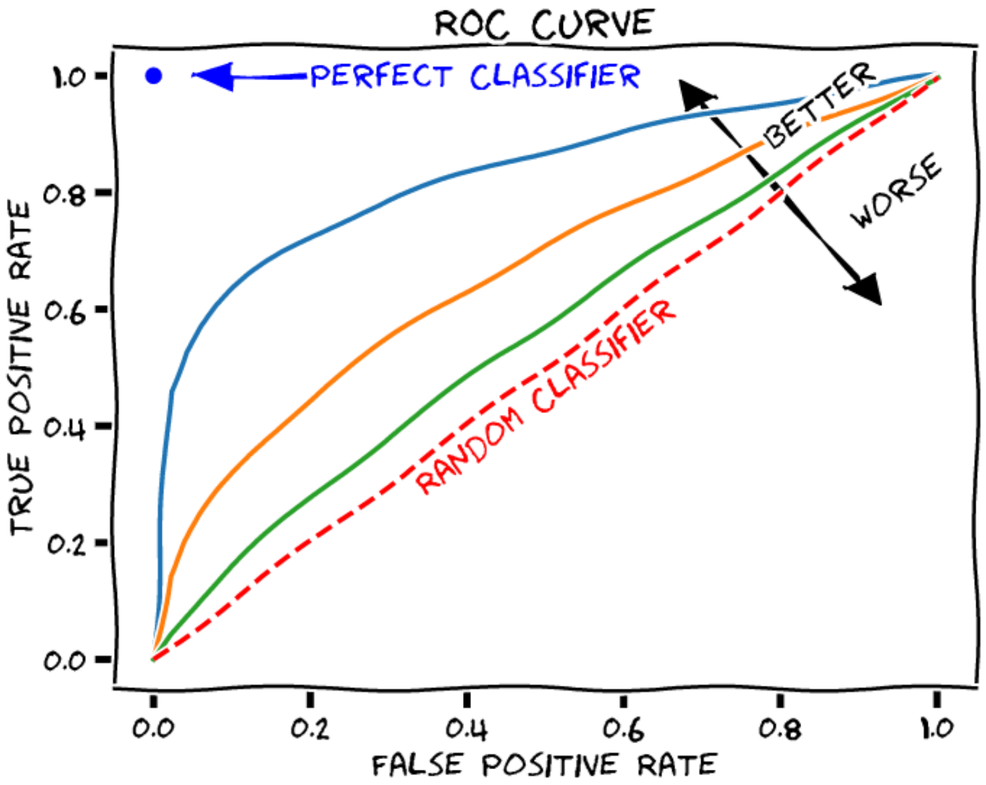
$$AUC = \sum_{i=1}^{n-1} (FPR_{i+1} - FPR_i) \times \frac{TPR_{i+1} + TPR_i}{2}$$

# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

- ROC Curves and AUC (Area Under the Curve)

| AUC Range | Interpretation |
|---|---|
| 0.9 – 1.0 | Excellent |
| 0.8 – 0.9 | Good |
| 0.7 – 0.8 | Fair |
| 0.6 – 0.7 | Poor |
| 0.5 – 0.6 | Fail (no discrimination) |



# MODEL EVALUATION

**ROC Curves and AUC (Area Under the Curve)**

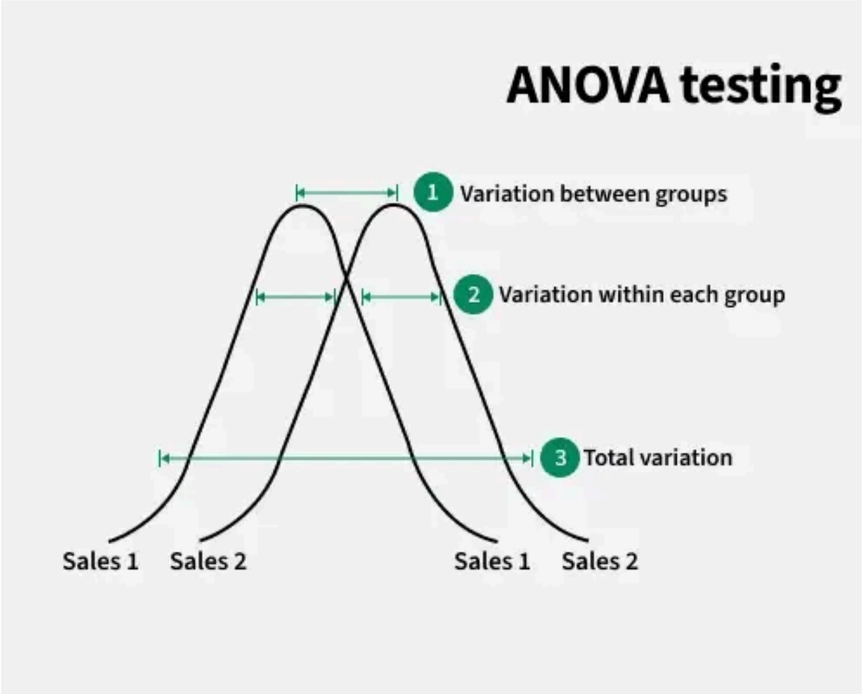
- ROC Curves and AUC (Area Under the Curve)

- The perfect model above, containing a square with sides of length 1, has an area under the curve (AUC) of 1.0. This means there is a 100% probability that the model will correctly rank a randomly chosen positive example higher than a randomly chosen negative example.
- In other words, looking at the spread of data points below, AUC gives the probability that the model will place a randomly chosen square to the right of a randomly chosen circle, independent of where the threshold is set.

# ANOVA analysis

ANOVA (Analysis of Variance) is a statistical method used to determine whether there are significant differences between the means of multiple groups.
In machine learning, ANOVA is primarily used for:

- Feature selection — to identify which independent variables (features) significantly influence the target variable.
- Model validation — to check whether the model's regression explains a significant portion of the variance.




**ANOVA helps us answer the question —**
**"Do these groups really differ, or are the differences just random?"**

# ANOVA analysis

**Core Idea**

ANOVA works by analyzing variability:

- Between-group variability: how much the group means differ from each other
- Within-group variability: how much the data points differ inside each group

If between-group variability is large compared to within-group variability, the group means are significantly different.

# ANOVA analysis

ANOVA uses a number called the F-value:

$$F = \frac{\text{Between-group variance (MSB)}}{\text{Within-group variance (MSW)}}$$

**If F ≈ 1, groups are similar (no significant difference).**
**If F > 1 and p-value < 0.05, at least one group differs significantly.**

$$MSB = \frac{SSB}{df_B}, \quad MSW = \frac{SSW}{df_W}$$

$$p = P(F_{df_B, df_W} > F_{calculated})$$

SSB (Sum of Squares Between) measures variation of group means around the overall mean
SSW (Sum of Squares Within) measures variation of individual scores around their group means
df_B = k - 1, df_W = N - k

(where k = number of groups, N = total number of observations)

# ANOVA analysis

| Type | Description |
|---|---|
| **Null Hypothesis ($H_0$)** | All group means are equal. There's no real difference. |
| **Alternative Hypothesis ($H_1$)** | At least one group mean is different. |

We use the p-value from the F-test to decide:
If $p < 0.05 \rightarrow$ Reject $H_0 \rightarrow$ There is a significant difference.
If $p \geq 0.05 \rightarrow$ Fail to reject $H_0 \rightarrow$ No significant difference.

# ANOVA analysis

**ANOVA is useful in production pipelines for model interpretability, feature analysis, and monitoring:**

**Feature Selection / Preprocessing**

Run ANOVA (or its equivalent F-test) to select numerical features that have statistically significant differences across target categories

In regression or scoring models, not all input variables truly explain the target. ANOVA helps determine which predictors significantly affect the output and which are just noise.

Use ANOVA to check whether the variance explained by each predictor is statistically significant ($p < 0.05$).

# ANOVA analysis

The residuals are differences between the observed output values and the fitted values:

$$R_i = y_i - f(x_i)$$

The size of the residuals, for all m samples in a data set, is related to the size of variance, and it can be estimated by

$$S^2 = \frac{\left[\sum_{i=1}^{m} (y_i - f(x_i))^2\right]}{(m-(n-1))}$$

These criteria are basic decision steps in the ANOVA algorithm, in which we analyze the influence of input variables on a final model. First, we start with all inputs and compute S2 for this model. Then, we omit inputs from the model one by one.

If we omit a useful input, the estimate S2 will significantly increase, but if we omit a redundant input, the estimate should not change much.

In principle, in each iteration, we compare two S2 values and analyze the differences between them. For this purpose, we introduce an F-ratio or F-statistic test in the form:

$$F = \frac{S^2_{\text{new}}}{S^2_{\text{old}}}$$

# ANOVA analysis

## ANOVA Analysis for a Data Set with Three Inputs X1, X2, and X3

| Case | Set of Inputs | $S_i^2$ | $F$ |
|---|---|---|---|
| 1 | $x_1, x_2, x_3$ | 3.56 | |
| 2 | $x_1, x_2$ | 3.98 | $F_{21} = 1.12$ |
| 3 | $x_1, x_3$ | 6.22 | $F_{31} = 1.75$ |
| 4 | $x_2, x_3$ | 8.34 | $F_{41} = 2.34$ |
| 5 | $x_1$ | 9.02 | $F_{52} = 2.27$ |
| 6 | $x_2$ | 9.89 | $F_{62} = 2.48$ |

Suppose that the data set has three input variables x1, x2, and x3 and one output Y.

In preparation for the use of the linear regression method, it is necessary to estimate the simplest model, in the sense of the number of required inputs.

Suppose that after applying the ANOVA methodology, the results given in Table are obtained.

The results of ANOVA analysis show that the input attribute x3 does not have an influence on the output estimation because the F-ratio value is close to 1

In all other cases, the subsets of inputs increase the F-ratio significantly, and therefore, there is no possibility of reducing the number of input dimensions further without influencing the quality of the model.

# Feature Significance

If p-value < 0.05, the predictor (here, region) has a real effect on satisfaction.
If p-value > 0.05, the variable is non-significant → remove it to simplify the model.

During feature selection: keep only statistically significant predictors.
For explainable AI reports: flag which business features (age, region, job type, etc.) have meaningful impact on model output.

Before training a model, we can quickly identify which features statistically relate to the target.

In Production
Automate in your feature-engineering pipeline:

- Run before model training.
- Drop or down-weight non-significant features.
- Reduces dimensionality and improves model interpretability.

# Detecting Overfitting

Overfitting means the model fits training data too well but generalizes poorly.
ANOVA helps by comparing model variance vs. residual variance across train/test (or time periods).

Perform ANOVA on model errors:

- Split predictions by dataset or time window (Train vs Test, or Month 1 vs Month 2).
- Compute F-test on residuals (prediction errors).

F very high and $p < 0.05$ → residuals differ significantly between train and test → overfitting likely.
$F \approx 1$ and $p > 0.05$ → no significant difference → model generalizes well.

Run this monthly to monitor stability of prediction errors.
Alert if test residual variance grows → retrain or regularize model.

# Detecting Overfitting and Drift

| Split Type | Example Groups |
|---|---|
| Dataset | Training, Validation, Test |
| Time Window | January, February, March |
| Segment | Region A, Region B, Region C |

If the F-test shows significance ($p < 0.05$), it means:
The model's residuals behave differently across datasets or time.

This is a signal of overfitting or drift.

Over time, model inputs or errors might shift — known as data or concept drift.
ANOVA can detect significant differences across time segments:

Interpretation
$p < 0.05$ → model error distributions differ significantly → drift detected.
Trigger data-drift monitoring alert → investigate data changes or retrain.

In Production
Include ANOVA test in monitoring dashboard (e.g., MLflow, EvidentlyAI).
Visualize F-statistic trend → detect when error variance spikes.

# Model Validation & Comparison

When you add new predictors or retrain, ANOVA can test whether the new model significantly improves fit compared to the old one.

Interpretation

- If $p < 0.05$, the full model adds significant explanatory power.
- If not → extra variables don't help and may overfit.

In Production

Use before deploying new model version:

- Is Model v2 statistically better than v1?
- Integrate in CI/CD or MLOps validation step.

# THANK YOU